

```
>daterange <-
colnames(df[,c(5:ncol(df))])
>countrynames <- df$Country.Region
>tdf <- data.frame(t(df),stringsAsFactors=FALSE)
Eliminate Province.State
>tdf <- tdf[c(-1),]
>rownames(tdf) <- daterange
>colnames(tdf) <- countrynames
>tdf$date <- as.Date(seq(as.Date("2020/1/22"), by = "day",length.out=nrow(tdf)),format = "%d/%b/%Y-%u")
>class(tdf$date)
>tstdf <- ts(tdf$China,frequency=365,
start=as.POSIXct(as.Date("2020/1/22")))
>plot.ts(tstdf, xlab="Country",
ylab="Count", main="Time Series graph")
>tstdf <- ts(tdf$India,frequency=365,
start=as.POSIXct(as.Date("2020/1/22")))
>plot.ts(tstdf, xlab="Country",
```

```
ylab="Count", main="Time Series graph")
```

Density distribution plot

Visualization of a variable depends on its type and corresponding graphical tools. For the variable 'country' I have already used a bar plot, so as to show the total infection of each category. To visualise the distribution of the total infections of each country, I can use a histogram and density plot using the *ggplot()* function.

Unlike the bar and trend analysis, here I have considered all the countries to have a better depiction of the density distribution. As mentioned above, the data is initially collected and filtered. The country-wise infection count is then done on country-wise group data. Finally, the *geom_histogram()* and *geom_density()* functions of *ggplot* are used for the histogram and density plot.

The density plot is overlaid on the histogram for a better understanding of the distribution. Moreover, a vertical line indicating the mean value is also shown.

```
#Preprocessing
>library(data.table)
> df <- fread("https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_19_time_series/time_series_covid19_confirmed_global.csv")
>df <- data.frame(df,
stringsAsFactors=FALSE)
>library(dplyr)
>df$Province.State <- sub("^$", NA, df$Province.State)
>df <- transform(df, Province.State = ifelse(Province.State==" " | is.na(Province.State), Country.Region, Province.State))
#Replace all fields with 0 if the value is NA
df <- df %>% replace(=="NA", 0) #
replace with 0
>dfCountryGroupSum <- aggregate(df[,c(5:ncol(df))],by=list(Category=df$Country.Region),FUN=sum)
>dfCountryGroupSum$Total <-
>apply(dfCountryGroupSum[,c(2:ncol(dfCountryGroupSum))],1,sum,na.omit=T)
>names(dfCountryGroupSum)
[1]<-c("Country")
>TotConfirmedCase <- dfCountryGroupSum[,c(1,ncol(dfCountryGroupSum))]
>names(TotConfirmedCase)[2] <- c("TotCase")
>p<-ggplot(TotConfirmedCase,
aes(x=log(TotCase))) +
  geom_histogram(aes(y=..density..),
colour="black", fill="white")+
  geom_density(alpha=.2, fill="#FF6666")
+
  geom_vline(aes(xintercept=mean(log(TotCase))),
color="blue",
linetype="dashed", size=1)+
  xlab("Country wise total Number of Case")
>p
```

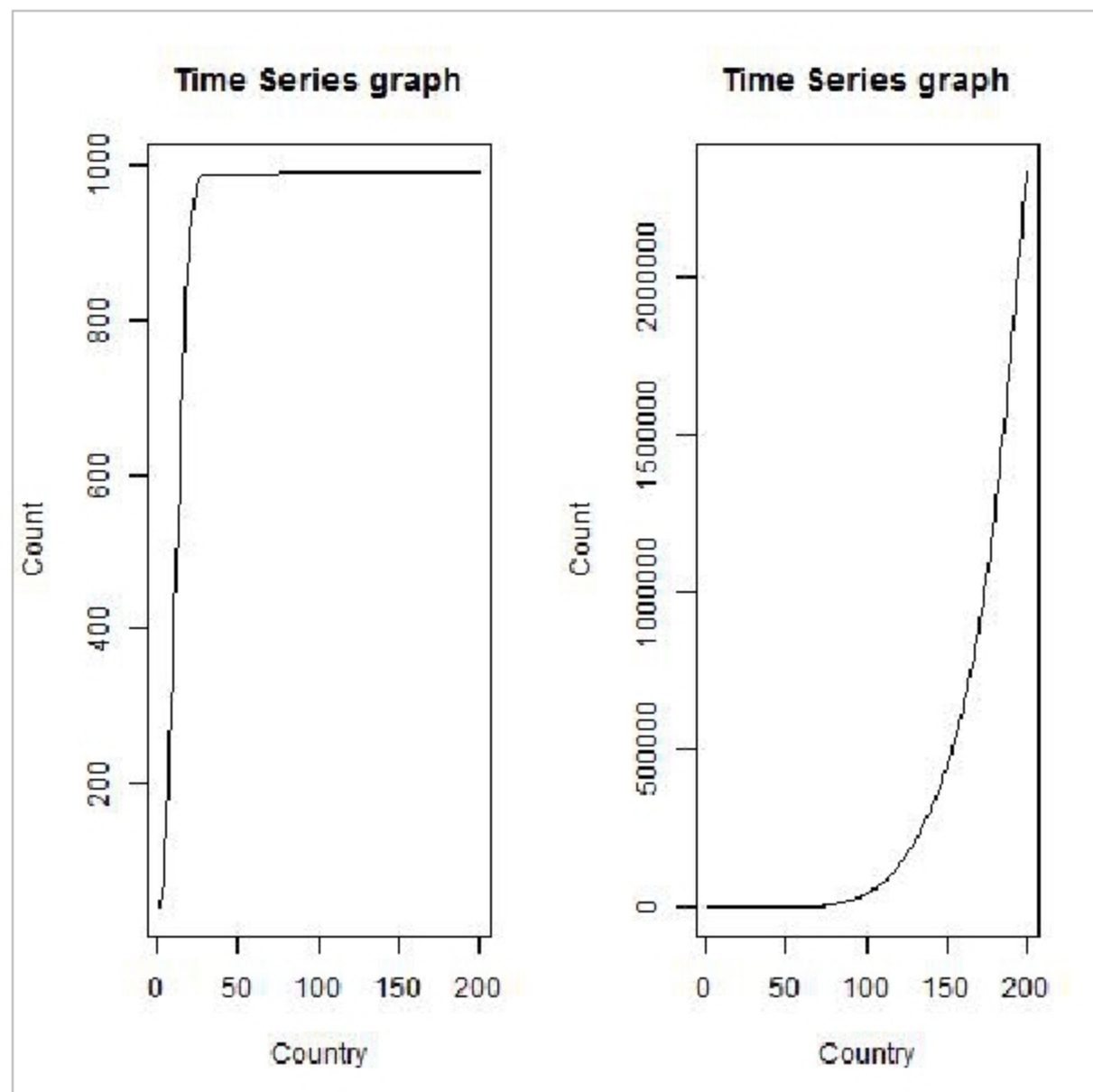


Figure 2: Trend analysis of COVID-19 infection of two countries